

# Cheng-Hao (Howard), Kao



## About me

A second-year MSc student in Mechatronics engineering at the University of Waterloo. Proficient in machine learning and data analysis. Possess the analytical abilities required for research and design as well as a comprehensive grasp of engineering principles. Currently researching on multi-agent system workloads as an intern at Huawei, and working in the Structural Biomechanics Lab on ML-based injury prediction. Previously involved in the Robotics and Computer Vision Lab conducting research on multi-robot collaboration joint trajectory planning and control systems and in the Microelectromechanical Systems Lab dedicated to integration of FPCB mirror-based scanners. Seeking more professional training for further development within the field of AI, data analysis, and cognitive science.

## Contact

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## Languages

Chinese - Native Language  
English - Professional Knowledge

## EDUCATION

January  
2024 -  
Present

**MSc in Mechanical and Mechatronics Engineering** 📍 Waterloo, On, Canada  
**University of Waterloo**  
*GPA: 3.90/4.0 Cumulative*

**Relevant Coursework:** Introduction to Machine Learning, Pattern Recognition and Machine Learning, Deployment of Deep Learning Model, Computational Intelligence, Optimization Methods, Human Factors in Testing

August 2018  
- June 2023

**Bachelor of Engineering** 📍 Toronto, On, Canada  
**Toronto Metropolitan University**  
*GPA: 3.85/4.33 Cumulative*

**Relevant Coursework:** Mechanical Design, Control of Robotic Manipulators, Real-Time Computer Control Systems, Fundamentals of Robotics, Mechatronics Systems Design, Measurements, Sensors and Instruments, Microprocessor Systems, Control Systems Digital Systems, Computer Programming Fundamentals, Electric Machines and Actuators, Electric Circuits, Applied Finite Elements, Vibrations, Machine Design, Mechanics of Machines, Solid Mechanics, Statics and Mechanics of Materials, Dynamics, Manufacturing Fundamentals, Introduction to Engineering Design, Engineering Graphical Communication, Numerical Analysis, Linear Algebra, Differential Equations and Vector Calculus, Calculus II, Calculus I

August 2015  
- June 2018

**High School Diploma** 📍 Shanghai, China  
**No.2 High School of East China Normal University**  
*Average: 95.3/100*

## WORK EXPERIENCE

August 2025  
- Present

**Technology Planning and Cooperation Research Intern** 📍 Waterloo, On, Canada  
*Huawei Canada*

- Multi-agent system workload trend identification
- Multi-agent system system-level optimization analysis.

May 2024 -  
April 2024

**Teaching Assistant** 📍 Waterloo, On, Canada  
*University of Waterloo*

- Teaching Circuits
- Teaching Mechanics of Deformable Solids
- Teaching Statics

June 2023 -  
January  
2024

**Research Assistant** 📍 Toronto, On, Canada  
*Toronto Metropolitan University RCVL Lab*

- Research on multi-robot collaboration joint trajectory planning and control
- Research on micro-plastic detection via ultrasonic sensor and machine learning

Professional Skills

Deep LearningMachine Learning

Data AnalysisOptimization

Signal ProcessingRoboticsControl Systems

MicrocontrollersSimulation and Design

Engineering Mathematics

Honors and Awards

Dean's List 2018 to 2019Dean's List 2019 to 2020

Dean's List 2020 to 2021Dean's List 2021 to 2022

Mechanical Engineering Alumni Award

Mechanical Engineering Capstone Project Award

Graduation with Distinction

Soft Skills and Strengths

AdaptabilityProblem SolvingCollaboration

Team WorkingOpen-Mindedness

Critical ThinkingCommunication

Willingness to LearnResponsibilityReliability

EnthusiasmAutonomyMotivated

Stress ToleranceResearch and Analysing

Personal Time ManagementStrategic Planning

PatienceEmpathyDiscipline

Interests

- Deep Learning
  - Machine Learning
  - Cognitive Science
  - Agentic AI
  - Control Systems
- Robotics
  - Billiards
  - Board Games
  - Card Games
  - Squash
  - Basketball

September 2022 - February 2023

**Research Assistant** Toronto, On, Canada  
*Toronto Metropolitan University MEMS Lab*

- Research on FPCB mirror-based lidar scanners
- Integration of PSD sensors, ToF sensors, and GUI

April 2021 - September 2021

**Research Assistant Intern** Toronto, On, Canada  
*Toronto Metropolitan University Ultrashort Laser Nanomanufacturing Lab*

- Conducting porosity and fiber analyses via SEM image processing
- Generating molecular dynamics simulations

April 2020 - September 2020

**Research Assistant Intern** Toronto, On, Canada  
*Toronto Metropolitan University Ultrashort Laser Nanomanufacturing Lab*

- Conducting porosity and fiber analyses via SEM image processing
- Generating molecular dynamics simulations

PUBLICATIONS

|                      |                                                                                                                                                          |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Manuscript 2025      | <b>Predicting the Risk of ACL Injuries Using Machine Learning Models and Anatomical Predictors</b> , <i>Under Review</i> , Link Unavailable              |
| Arxiv Preprint 2024  | <b>Morphological Detection and Classification of Microplastics and Nanoplastics Emerged from Consumer Products by Deep Learning</b> , <i>IEEE</i> , Link |
| Journal Article 2023 | <b>MR. CAP: Multi-Robot Joint Control and Planning for Object Transport</b> , <i>IEEE Control System Letters</i> , Link                                  |

SOFTWARE SKILLS

|                         |                                                 |                                                     |
|-------------------------|-------------------------------------------------|-----------------------------------------------------|
| Data Analysis           | PyTorch<br>AutoGluon<br>XGBoost<br>SAM<br>NLopt | TensorFlow<br>SkLearn<br>CATBoost<br>YOLO<br>Optuna |
| Robotics                | GTSAM                                           |                                                     |
| Modeling and Simulation | Simulink<br>ANSYS                               | Multisim                                            |
| Design and Development  | SolidWorks<br>LabVIEW                           | Matlab                                              |
| Version Control         | GitHub                                          |                                                     |
| Office Automation       | MS Office (Excel, Word, PowerPoint)<br>Overleaf | Adobe Illustrator                                   |

PROGRAMMING LANGUAGES

- Python
  - Matlab
  - Processing
  - C++
- C
  - LaTex
  - Fortran